

Switching → between 2 machines inside a LAN using their MAC addresses

Routing → Machines across LANs using their IP addresses.

Routers have to keep routing table. For a IP packet to guide in right direction based on dest IP address.

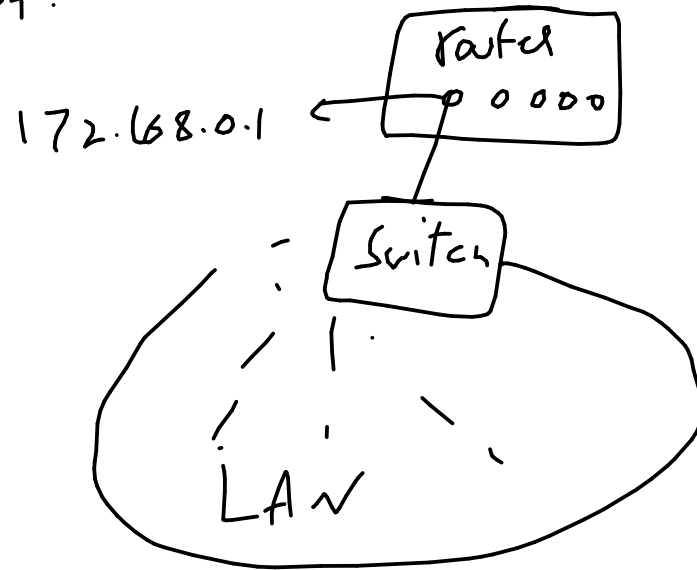
→ To keep routing tables efficient, we don't store individual IP addresses and which port to go to.

→ we assign a subnet to a LAN and we send to the lan.

→ Just have to store path for subnet, not every IP

On a Router, How a LAN is Connected.

- First Construct your whole LAN with the one main/core Switch.
- Connect one port in Switch to one port in router.
- ⇒ Now your whole LAN includes all switches, PCs, end devices and the Connecting Port in router. → your gateway basically.
- Now allocate whatever Subnet of IP addresses to your LAN. every device will take one IP from it. along with your gateway port.



if Subnet for LAN
is
 $172.168.10.0/24$
your router
Port / gateway uses
one IP from this.

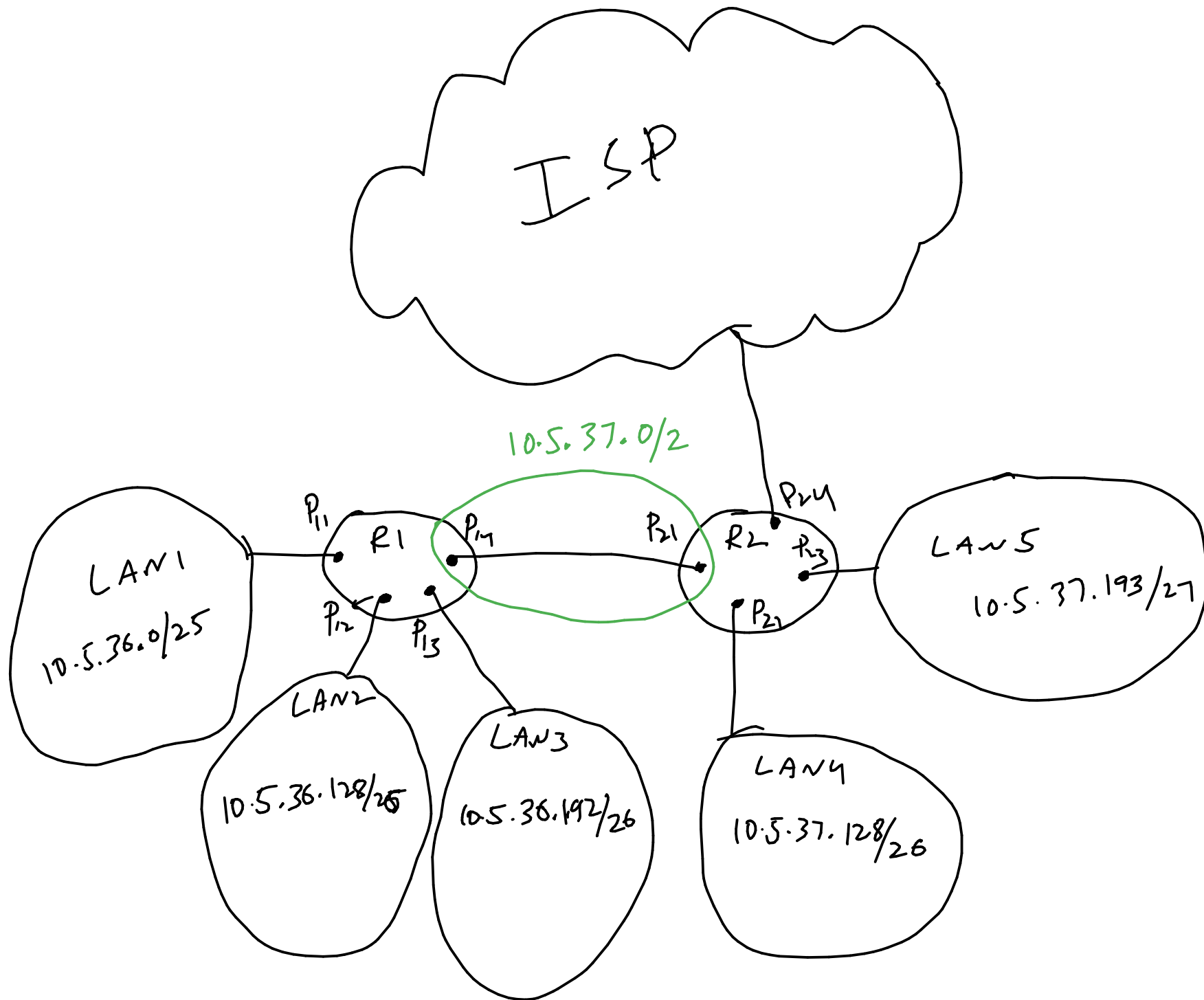
Routers route traffic across LANs

a routing table should contain direction for every IP address
(Ideally)
→ we can do like discard some LANs for which
we don't want to send anything.

Routing table records

< Subnet of IP addresses, which Port in the router to forward >

the green circle
is also considered
a LAN and
they also need
a small subnet
of IPs.



Routing table of R1

10.5.36.0/25 — P₁₁

10.5.36.128/26 — P₁₂

10.5.36.192/26 — P₁₃

default — P₁₄

R2

10.5.37.128/26 → P₂₂

10.5.37.193/27 → P₂₃

10.5.37.0/2

10.5.36.0/25

10.5.36.128/26

10.5.36.192/26

default

→ P₂₁

→ P₂₄

we can further
compress to

10.5.37.0/2

10.5.36.0/24

} → P₂₁ →

just forward through P₂₁ to

Router 1 → R1 table

decides where to go.

Think of world like